

Title

Integrating digital technology with direct patient support to improve TB medication adherence: protocol for a pragmatic randomized trial

Abstract

Tuberculosis remains a leading infectious cause of mortality globally, with medication non-adherence driving both treatment failure and drug resistance. Digital tools offer scalable adjuncts to clinical supervision, yet real-world implementation barriers remain incompletely understood. The TB Treatment Support Tools (TB-TST) intervention combines a smartphone application delivering personalized medication reminders, educational content, and bidirectional communication with a reliable, objective adherence biomarker (urine drug metabolite assay). This pragmatic randomized trial protocol describes evaluation of TB-TST integrated with standard directly observed therapy in four urban TB treatment centers in Argentina. A total of 90 newly diagnosed TB patients will be randomized 1:1 to standard care or standard care plus TB-TST. Primary outcomes are treatment completion rates and medication adherence measured by metabolite testing at predetermined intervals. Secondary outcomes include patient satisfaction, technical usability, cost-effectiveness, and qualitative insights into implementation barriers and facilitators. This study aims to generate evidence for scalable, patient-centered digital adherence strategies in TB programs.